

KARUNA VASU

Software Engineer

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PROFESSIONAL SUMMARY

Software Engineer building automation and AI systems for US Healthcare RCM. Delivered ~95% faster claim processing and 40–80% less manual work. Core work spans App Hub / BotVeta orchestration, Automation Productivity Dashboard (tracking 371 projects, ~118K jobs/week, ~\$20M charge-cost visibility), Azure OpenAI PDF extraction, Slack SLA operations, and AWS IVR automation.

SKILLS

Languages: Python, SQL, JavaScript, TypeScript, C#, PERL, R

Backend & APIs: FastAPI, REST APIs, microservices, ASP.NET Core (.NET 9), Hangfire, Flask, Django, SQLAlchemy, JWT RBAC

Frontend: React, Next.js, Vite, Tailwind CSS

Automation: Selenium, RPA, workflow automation, Availity RCM pipelines

AI/ML: Azure OpenAI, Amazon Bedrock (Claude 3.5), OCR (RapidOCR), LLMs, prompt engineering, document AI

Data Processing: Pandas, NumPy, rule engines, data validation, openpyxl

Cloud & DevOps: AWS (Connect, Bedrock, Lambda, EC2, DynamoDB, IAM), Docker, CI/CD, Azure Key Vault, Azure DevOps

Databases: PostgreSQL, Microsoft SQL Server, SQL, SQLite, vector DB (basic)

Platform & Tooling: APScheduler, Serilog, PyODBC, SFTP automation, Git, Linux, Postman, Jenkins, Kubernetes, Power Automate

EXPERIENCE

Software Engineer

Integrity Healthcare Solutions Pvt Ltd, Ahmedabad | December 18, 2023 – Present

- Reduced claim turnaround by ~95% with Python automation pipelines integrated with Availity and RCM workflows.
- Automated eligibility, authorization, claims, and EOB/ERA flows using Python and Selenium; cut manual effort by 40–80%.
- Built FastAPI microservices for high-volume production automation with retries and SQL Server logging.
- Architected BotVeta / App Hub: Hangfire schedules, Audit Center, auth rate limits on login/OTP, product-usage analytics, and Product Weekly Update (UTC date ranges + comparison).
- Built Automation Productivity Dashboard tracking 371 ADO projects (PM/SR CSM, discussions + status/color), ~118K jobs/week (84% success), and 2,770 FTE sent (~\$20M charge-cost visibility).
- Shipped AI PDF Extractor (Azure OpenAI + OCR): Appointment List / HCO tracker / Trust-PHNT-NBA templates with validated saved_minutes → FTE KPIs.
- Delivered Slack Thread Tracker: auto-resolve image-only messages, reaction-driven status, pre-shift digests, AI triage, and SLA analytics.
- Engineered IVR automation on Amazon Connect + AWS Bedrock for payer calls; reduced manual calling effort by 60–80%.

Data Science Intern

Zephyr Technologies and Solutions Pvt Ltd, Mangalore | June 2023 – July 2023

- Developed a real-time chat application using Django, WebSockets, and PostgreSQL.

Research Intern

Manipal Institute of Technology, Manipal | February 2023 – December 2023

- Published IEEE research paper on ECG signal analysis; awarded Best Paper & Best Presenter (NMITCON 2023). DOI: <https://doi.org/10.1109/NMITCON58196.2023.10276059>

EDUCATION

Master of Science: Bioinformatics — Manipal Academy of Higher Education, Manipal | 2021 – 2023 | GPA: 7.24

Bachelor of Science: Bioinformatics — Sardar Patel University, Anand | 2018 – 2021 | GPA: 7.00

KEY PROJECTS

- **BotVeta / App Hub** — schedules, Audit Center, auth rate limits, product-usage, Product Weekly Update.
- **Automation Dashboard** — 371 projects · PM/SR CSM · ~118K jobs/week · ~\$20M.
- **AI PDF Extractor** — Appointment List / HCO / Trust-PHNT-NBA; saved_minutes → FTE.
- **Slack Thread Tracker** — image-only auto-resolve, reactions, digests, SLA.
- **IVR / RCM Suite** — Connect+Bedrock; EV/Auth/Claims automation.